NEET Important Chapter - Motion in a Straight Line

Important Concept of Motion in a Straight Line

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In this chapter we will talk about the motion of any object or a body in a straight line or in 1 dimension. In this chapter we will also understand the meaning of motion and what is the state of rest and state of motion of an object. We have covered all the important parameters that are required to understand the motion in 1-D.

In this article, we have provided all the important concepts like path length, distance, displacement, velocity, and acceleration then formula for determining them. We have also discussed the graph between distance v/s time, velocity v/s time, and acceleration v/s time. Students will also get to know about free fall and motion under gravity.

This chapter also deals with the kinematic equations that are very important for solving any kind of problem in one dimensional motion. Motion in a straight line is also known as rectilinear motion.

Now, let's move onto the important concepts and formulae related to NEET exams along with a few solved examples and previous year questions.

Important Topics of Motion in a Straight Line

- Motion in a straight line notes
- NCERT Solution of motion in a straight line
- Velocity
- Acceleration
- Motion in a straight line formulas
- Uniform Motion and Non Uniform Motion
- Distance and Displacement
- Instantaneous Speed and Velocity
- Kinematics Equations

Motion in a Straight line Important Concept for NEET

Name of the Concept	Key Points of the Concepts
1. State of rest	• Object is said to be at a state of rest when it does not change its position with respect to the time.
2. State of Motion	• The body or object is said to be in a state of motion with respect to the other body if it changes its position with respect to the time.

3. Rectilinear or linear Motion	• When a body or object moves in such a way that it covers only linear distance then it is said to be a linear motion or translational motion. And it is also known as one dimensional motion. • Under this motion, Only one coordinate in the position is changed. • Ex. A ball falling freely under gravity.
4. Distance or Path length	• The distance is the total length of actual path traversed by the particle or a body in a definite interval of time. • distance is a Scalar quantity. • Its SI unit is meter. • It depends on the path, not on the final or initial positions of the particle.
5. Displacement	• Displacement is the minimum distance between the initial and final position of a particle in a given interval of time. • It is a vector quantity. • Its SI unit is meter.

6. Speed	• Speed is the rate of change of distance which is covered by the body or a particle with respect to time. • Speed can be classified as uniform and nonuniform. • Speed is a Scalar quantity • It's SI unit is m/s • **Uniform Speed -** when a particle covers equal distances in equal intervals of time then it is said to be moving with uniform speed. • **Non Uniform Speed** - In non uniform speed particles cover unequal distances in equal intervals of time or distance traveled by a particle is different in equal interval of time.

	• **Average Speed** - the average speed of a particle for a given interval of time is defined as the ratio of total distance traveled to a total time taken. • **Instantaneous Speed** - it is the speed of a particle at a particular instant of time.
7. Velocity	• Velocity is defined as the rate of displacement of a moving particle with respect to time. • It is a vector quantity. • Its SI unit is meters per second. • Velocity can be classified as uniform and nonuniform velocity. • Velocity may be positive, negative or zero. • **Uniform Velocity -** a particle is set to have a uniform velocity if magnitude as well as direction of

	its velocity remains same. This is possible only when it moves in the straight line without reversing its direction. • **Non Uniform Velocity -** A particle is said to have non uniform velocity if both either magnitude or direction of velocity changes. • **Average Velocity -** it is defined as the ratio of displacement to time taken by a particle and its direction is along the displacement.

8. Acceleration	• The rate of change of velocity of an object is called acceleration of the object. • It is a vector quantity. • It's direction is same as that of change in velocity (not in the direction of the velocity) • Its unit is — • **Uniform Acceleration** - A body said to have uniform acceleration if magnitude and direction of the acceleration remains constant

	during motion of the particle. • **Non Uniform Acceleration -** A body said to have non uniform acceleration if either magnitude or direction or both change during the motion. • **Average Acceleration -** it is the ratio of total change in velocity to the total taken time by the particle or a body. • **Instantaneous Acceleration -** it is the acceleration of a particle at a particular instant of time.

9. Free fall or Motion under gravity	• In the absence of air it is found that all bodies fall with the same acceleration near the surface of earth this motion of body falling towards the earth is called motion under gravity or free fall in free for acceleration of body is equal to acceleration due to gravity
10. Acceleration due to gravity	• Acceleration produced in a body by a force of gravity is called acceleration due to gravity. It is denoted by g. • Value of g = 9.8 —

List of Important formulas for Motion in a Straight line

S.No.	Name of the Concept	Formula
1.	Displacement	
2.	Average velocity	
3.	Instantaneous velocity	
4.	Average acceleration	
5.	Instantaneous acceleration	

6.	First equation of motion	Here u is initial velocity, v is a final velocity, a is acceleration and t is a time taken.
7.	Second equation of motion	— Here s is displacement of body, u is initial velocity, v is a final velocity, a is acceleration and t is a time taken.
8.	Third equation of motion	Here s is displacement of body, u is initial velocity, v is a final velocity and a is acceleration of a body.
9.	Displacement in nth second	—
10	Maximum height attained by a particle which is projected vertically upward with an initial velocity u	—

Solved Examples

1. A person travels along a straight road for the first half time with a velocity and the next half time with a velocity . The mean velocity of the man is

Sol:

Given person move some distance with velocity in half time let it is t second
Distance traveled in this time =
In the second half time he traveled distance with speed

Distance traveled in this time =

Total time = t + t = 2t

Total distance =

Mean velocity = —

Therefore, The mean velocity of the man is .

Key Point - For mean or average, we always take total distance and total time for the whole journey.

2. A body falls freely from rest. It covers as much distance in the last second of its motion as covered in the first three seconds. The body has fallen for a time of ____ second ?

Sol:

By newton's second equation of motion -

Displacement covered in nth second

Initial velocity is zero u = 0

2n = 10

n = **5 second**

Thus, The body has fallen for a time of 5 seconds.

Key Point: To solve this type of problem apply the formula of distance at a particular instant of time.

Previous Year Questions From NEET Paper

1. A ball is vertically downward with the velocity of 20 m/s from the top of a tower.

It hits the ground after some time with the velocity of 80 m/s. The height of the tower is:(g = 10 —) (NEET 2020)

Sol:

Given u = 20 m/s, v = 80 m/s

By using newton third equation of motion

By putting all values-

height of the tower = h = 300 m

Thus, we got the height of the tower h is 300 m.

Trick - Whenever a body is thrown downward then the acceleration is considered an acceleration due to gravity that is g. Therefore by applying the third equation of motion we can determine the height.

2. A car starts from rest and accelerates at 5 — at t = 4 s, a ball is dropped out of a window by a person sitting in the car. What is the velocity and acceleration of the ball at t = 6 s? (take g = 10 —) (NEET 2021)

Sol:

Given- initial velocity = u = 0

Velocity at t = 4 second -

V = u + at

Put all the values -

v = 0 + 5 (4) = 20 m/s

At t = 6 second acceleration due to gravity - a = g = **10** —

= 20 m/s (due to car)

= u + at = 0 + 5 (4) = 20 m/s

—

‾ **m/s**

Trick - upon going through the question and understanding it, we can first determine the acceleration at 6 sec and for final velocity we have to consider velocity in both x and y direction.

Practice Questions

1. A ball is thrown upward from the top of a tower 40 m high with the velocity of 10 m/s. Find the time when it is strikes the ground (**Ans: 4 sec**)

2. A Pebble is thrown vertically upward from a bridge with an initial velocity of 4.9 m/s. it strikes the water after 2 s. If acceleration due to gravity is 9.8 —

a. What is the height of the bridge?

b. with what velocity does the pebbler the water?(**Ans: 9.8m, 14.7 m/s**)

Conclusion:

In this article we have provided important information regarding the chapter Motion in a Straight line such as important topics, concepts, and formulae, etc.. Students should work on more solved examples for securing good marks in the NEET exams.